V.L Memorial Public School

475 Vijay Nagar

CHAPTER-9-QUESTIONS-CLASS8TH-SCIENCE

Class 08 - Science (Curiosity)

Section A

1. Why does every sip of an ORS solution taste consistent? **[1]**

   a) Different components rise to the surface
   b) The components evenly distributed throughout the solution
   c) The components separate after preparation
   d) It contains only one ingredient

2. How can you identify a non-uniform mixture? **[1]**

   a) By appearance only
   b) By clear separation over time
   c) By inconsistency in component distribution
   d) By taste difference

3. Which of the following best describes why more oxygen can dissolve in cold water? **[1]**

   a) Cold water evaporates more slowly
   b) Cold water has fewer impurities
   c) Low temperature allows more gas molecules to remain dissolved
   d) Cold water has less salinity

4. What is the maximum amount of solute that dissolves in a fixed quantity of solvent at a given temperature called? **[1]**

   a) Dilution
   b) Solubility
   c) Saturation
   d) Concentration

5. Why can two objects of the same size feel different in heaviness? **[1]**

   a) They have different colours
   b) They have different temperatures
   c) They have different densities
   d) They have different shapes

6. What is the role of a solvent in a solution? **[1]**

   a) To create a heterogeneous mixture
   b) To dissolve another substance
   c) To separate substances
   d) To be dissolved

7. What happens to the solubility of gases when the temperature of the liquid increases? **[1]**

   a) It decreases
   b) It increases initially then decreases
   c) It increases
   d) It remains unchanged

8. What distinguishes a solution from a suspension? **[1]**

   a) Suspension has large particles that
   b) Solution settles quickly

settle

c) Both form clear liquids

d) Solutions and suspensions behave similarly

9. What is the liquid component called when a solid dissolves in it to form a solution? **[1]**

a) The solution

b) The solvent

c) The mixture

d) The solute

10. Why does stirring and heating allow more solute to dissolve in a solvent? **[1]**

a) It decreases the solvent particles

b) It decreases the solubility limits

c) It increases the energy available to solute particles, helping them integrate into the solution

d) It increases the solute particles

11. What happens when oil is added to water? **[1]**

a) Oil floats above water

b) Oil disappears into water

c) Oil sinks below water

d) Oil mixes throughout water

12. Which of the following is an example of a solid solution? **[1]**

a) Carbon dioxide dissolved in water

b) Salt dissolved in water

c) Gold dissolved in copper

d) Sugar dissolved in water

13. What happens to the solubility of most substances when temperature increases? **[1]**

a) Solubility remains unchanged

b) Solubility decreases

c) Solubility increases

d) Solubility initially increases and then decreases

14. Why does sawdust float while sand sinks when both are placed in water? **[1]**

a) Their densities differ

b) Their shapes differ

c) Their temperatures differ

d) Their colours differ

15. Which factor primarily affects the density of gases? **[1]**

a) Pressure

b) Shape

c) Size

d) Temperature

16. What type of mixture is formed when gases dissolve in water? **[1]**

a) Uniform mixture

b) None of these

c) Non-uniform mixture

d) Heterogeneous mixture

17. Which substance acts as the solvent in air? **[1]**

a) Nitrogen

b) Argon

c) Carbon dioxide

d) Oxygen

18. Why does ice float on water? **[1]**

a) Ice is denser than water, causing it to float.
b) Ice floats due to the effect of Earth's gravity.
c) Ice floats because it is lighter than air.
d) Ice is less dense than water, causing it to float.

19. What is the solid component called when it dissolves in a liquid to form a solution? **[1]**

a) The solution
b) The mixture
c) The solvent
d) The solute

20. Which type of water generally contains more dissolved oxygen? **[1]**

a) Cold water
b) Warm water
c) Heated water
d) Boiling water

21. **Assertion (A):** If you add an iron rod to water, it will sink. **[1]**
Reason (R): Iron has a higher density than water.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** Mixing sugar in water creates a solution. **[1]**
Reason (R): In solutions, the liquid is always the solute.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** Rice grains settle at the bottom of the water container. **[1]**
Reason (R): Rice is denser than water.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Chalk powder does not dissolve in water. **[1]**
Reason (R): Chalk powder and water form a uniform mixture.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

25. **Assertion (A):** Dissolved oxygen in water is crucial for aquatic life. **[1]**
Reason (R): Oxygen is present in water in very high concentrations.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

26. Why is the mixture of gases in water considered a uniform mixture? **[1]**

27. What makes a solution dilute as compared to another? **[1]**

28. An object has a mass of 400 g and a volume of 40 cm^3. What is its density? **[1]**

29. Determine how the solvent is identified in liquid-liquid solutions. **[1]**

30. A stone sculpture weighs 225 g and has a volume of 90 cm^3. Calculate its density and predict whether it will float or sink in water. **[1]**

31. Evaluate why a gaseous mixture like air is considered a solution. **[1]**

32. Calculate the density of a stone with mass 50 g and volume 20 cm^3. **[1]**

Section B

33. Describe how oxygen in water affects aquatic organisms. **[2]**

34. Define the term 'solute' and 'solvent' in the context of solutions. **[2]**

35. How does heat affect the density of gases? **[2]**

36. What is the SI unit of density, and how is it determined? **[2]**

37. Classify the components of air as solutes or solvent. **[2]**

38. Explain the difference between a dilute solution and a concentrated solution with examples. **[2]**

39. Differentiate between unsaturated and saturated solutions. **[2]**

Section C

40. Explain how repeated addition of salt demonstrates solubility and saturation. **[3]**

41. Why do hot air balloons rise? Use the idea of density in the answer. **[3]**

42. What are the characteristics of a uniform mixture? **[3]**

43. Explain why understanding density is important in the design of ships and submarines. **[3]**

44. Under what conditions is the density of a substance affected by temperature and pressure? **[3]**

45. What is meant by the solubility of gases in water? Give examples. **[3]**

46. Explain why bamboo and wooden logs were suitable materials for traditional rafts and boats. **[3]**

Section D

47. **Read the following text carefully and answer the questions that follow:** **[4]**

Many gases, such as oxygen, dissolve in water to a small extent. This dissolved oxygen is essential for aquatic life. The solubility of gases decreases as temperature rises; hence, more oxygen dissolves in cold water than in warm water. Mixtures can be uniform (solutions) where components are not separately visible, or non-uniform, where components are visible, like sand in water that sinks, or sawdust in water that floats.

Questions:

a. What type of mixture is formed when gases dissolve evenly in water? **(1)**

b. Why is dissolved oxygen in water important for aquatic life? **(1)**

c. How does temperature affect the solubility of oxygen in water? **(2)**

OR

Differentiate between uniform and non-uniform mixtures with examples. **(2)**

48. **Read the following text carefully and answer the questions that follow:** **[4]**

Density is the mass per unit volume of a substance and does not depend on its shape or size, but varies with temperature and pressure. Heating generally decreases density as particles spread apart, while cooling increases it. Pressure mainly affects gases by

compressing particles and increasing density, but it has negligible effects on solids and liquids. Water shows unusual behavior: ice floats because it expands on freezing, lowering its density compared to liquid water.

Questions:

a. Which factor has the greatest effect on the density of gases? **(1)**

b. Why does ice float on water even though both are the same substance? **(1)**

c. How does temperature affect the density of substances? **(2)**

OR

Why does pressure have a significant effect on gases but negligible effect on solids and liquids? **(2)**

49. **Read the following text carefully and answer the questions that follow:** **[4]**

Some objects float while others sink in water. For example, husk floats while rice sinks. Similarly, oil floats on water. People often think lighter objects float and heavier ones sink. However, this depends on a property called density, which is the heaviness of a substance. A wooden stick and an iron rod of the same size differ because iron is denser. But density is not the only factor that determines floating or sinking.

Questions:

a. Which property mainly explains why some objects float while others sink? **(1)**

b. Why does a wooden stick float while an iron rod of the same size sinks? **(1)**

c. Explain with examples how density affects whether an object floats or sinks. **(2)**

OR

Why is it incorrect to say that only heavier objects sink in water? **(2)**

Section E

50. Explain why two objects of the same size, such as a wooden stick and an iron rod, behave differently when placed in water. **[5]**

51. Evaluate the significance of solvents in the scientific heritage described, including their role in medicinal formulations and Asima Chatterjee's work. **[5]**

52. Discuss how temperature affects the solubility of gases in liquids, specifically regarding dissolved oxygen in water. **[5]**

53. Why do gases dissolve uniformly in water, forming a uniform mixture rather than a non-uniform one? **[5]**

54. Define unsaturated and saturated solutions in the context of the solubility limit of a solute. How does the state of the solution change as more solute is added to the solvent? **[5]**

55. Explain the concept of relative density and how it can be used to compare substances. **[5]**

Section F

56. **State whether the given statement is True or False:** **[10]**

(a) Density is the only factor that determines whether an object will float or sink in a liquid. **[1]**

(b) A wooden stick floats on water because it has a higher density than water. **[1]**

(c) The amount of space occupied by any object is called its mass. **[1]**

(d) Rice sinks in water because it is denser than water. **[1]**

(e) Objects that sink in water are always heavier than the objects that float. **[1]**

(f) Oxygen gas is more soluble in hot water rather than in cold water. **[1]**

(g) A solution is a homogeneous mixture. **[1]**

(h) An Oral Rehydration Solution (ORS) is used to treat dehydration. **[1]**

(i) Density is independent of both temperature and pressure. **[1]**

(j) The solubility of most solids increases with an increase in temperature. **[1]**

57. **Fill in the blanks:** **[10]**

(a) Water is often called the universal _____ because it can dissolve many substances. **[1]**

(b) An Oral Rehydration Solution (ORS) is used to treat _____. **[1]**

(c) The particles in a solution do not _____ a beam of light passing through it. **[1]**

(d) When a solute is dissolved in water, the solution is called an _____ solution. **[1]**

(e) Oxygen dissolves only in _____ quantities in water, yet it is vital for aquatic life. **[1]**

(f) The formula used to determine the density of a substance is Density = _____. **[1]**

(g) The solution in which glucose has completely dissolved in water, and no more glucose can dissolve at a given temperature, is called a _____ solution of glucose. **[1]**

(h) In colder conditions, the density of a substance usually _____. **[1]**

(i) The maximum amount of solute that can dissolve in a fixed quantity of solvent at a given temperature is called _____. **[1]**

(j) A solution is a _____ mixture in which components are evenly distributed. **[1]**